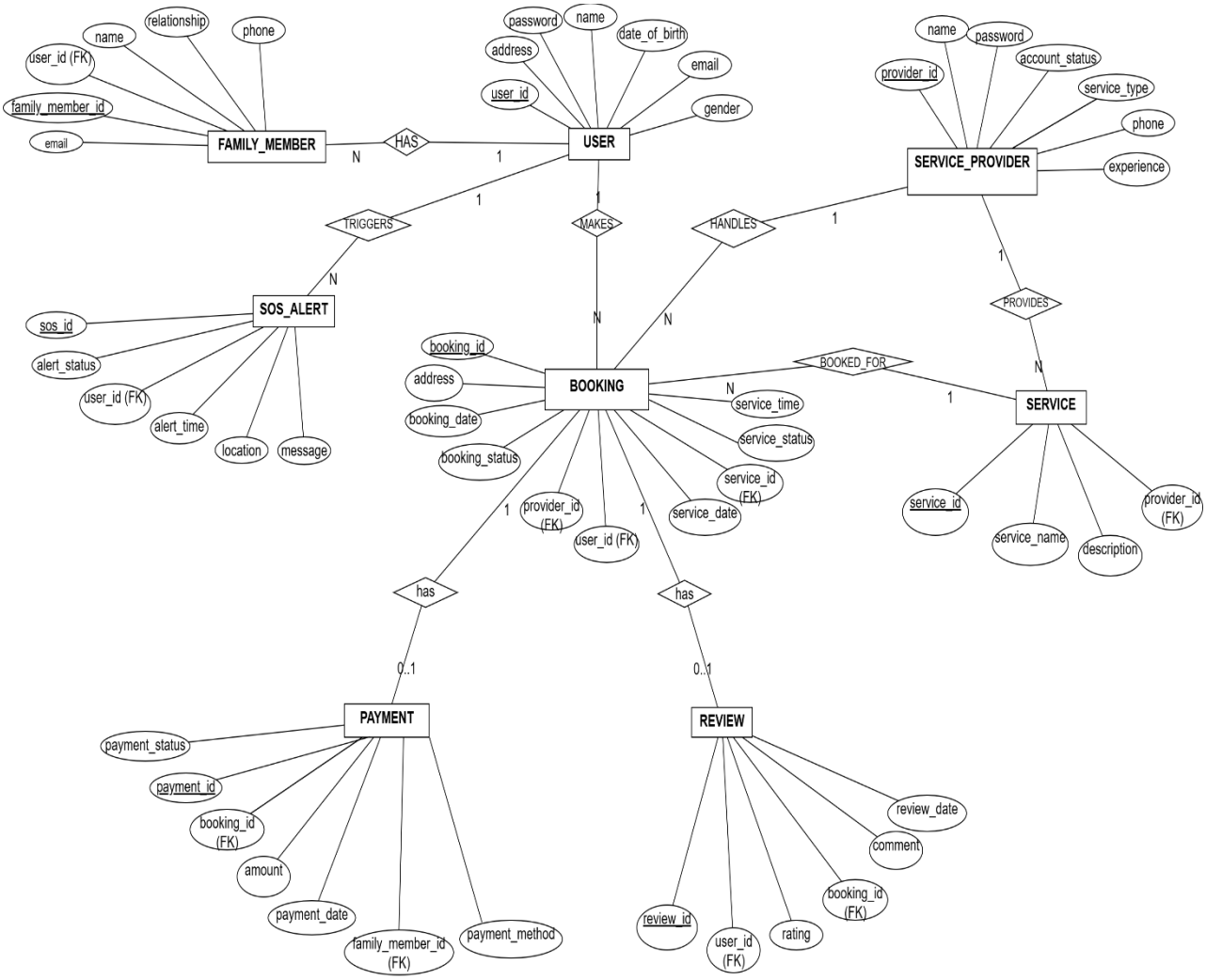
ER Diagram of CareConnect

Smart Elder Assistance and Trusted Service provider Platform

ER Diagram is known as Entity-Relationship Diagram. it is used to analyze the structure of the database. It shows relationships between entities and their attributes. An ER Model provides a means of communication. The ER diagram of CareConnect is derived from the platform's database design and has the following description:

1. A user registers with personal information such as name, email, password, phone, address, date of birth, gender and account status.
2. A user may have several family members who are identified by name, relationship, phone, email and user reference.
3. A user can make multiple bookings for available services.
4. Service providers offer services and are identified by provider information, service type, experience, availability and account status.
5. A booking stores booking date, service date, service time, address and booking status.
6. A booking may have payment and review information.
7. Users can trigger SOS alerts containing alert time, location, message and alert status.

ER Diagram of CareConnect



Entities and their Attributes

Users Entity : Attributes are user_id, name, email, password, phone, address, date_of_birth, gender and account_status. user_id is the Primary Key.

Family_Member Entity : Attributes are family_member_id, name, relationship, phone, email and user_id. family_member_id is the Primary Key and user_id is a Foreign Key.

Service_Provider Entity : Attributes are provider_id, name, password, email, account_status, service_type, phone and experience. provider_id is the Primary Key.

Booking Entity : Attributes are booking_id, address, booking_date, booking_status, service_date, service_time, service_id, user_id and provider_id. booking_id is the Primary Key; service_id, user_id and provider_id are Foreign Keys.

Service Entity : Attributes are service_id, service_name, description, and provider_id. service_id is the Primary Key and provider_id is a Foreign Key.

SOS_Alert Entity : Attributes are sos_id, alert_status, user_id, alert_time, location and message. sos_id is the Primary Key and user_id is a Foreign Key.
Payment Entity : Attributes are payment_status, payment_id, booking_id, amount, payment_date, family_member_id and payment_method. payment_id is the Primary Key; booking_id and family_member_id are Foreign Keys.
Review Entity : Attributes are review_date, comment, booking_id, review_id, user_id and rating. review_id is the Primary Key; booking_id and user_id are Foreign Keys.

Relationships

- Users has Family_Members 1:N : One user can have several family members but each family member belongs to one user.
- Family_Member triggers SOS_Alert 1:N : One family member can trigger multiple SOS alerts but each SOS alert is associated with one family member.
- Users makes Booking 1:N : One user can make many bookings but each booking belongs to one user.
- Service_Provider provides Service 1:N : One service provider can provide several services but each service is provided by one provider.
- Booking handles Service N:1 : Many bookings can be associated with one service but each booking is associated with one service.
- Booking is booked for Service N:1 : Many bookings can be booked for one service but each booking is booked for one service.
- Booking has Payment 1:0..1 : A booking may have one payment or no payment.
- Booking has Review 1:0..1 : A booking may have one review or no review.